

Intro to Linear Algebra for Machine Learning

* Matrices

What is matrix?



pure white

0 → 255

element

pure black

1MP

$A =$
 2×2

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

rows

2×2

Dimension

one data
Structure

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} a-3 & b-4 \\ c+d & c-d \end{bmatrix}$$

a_{00}

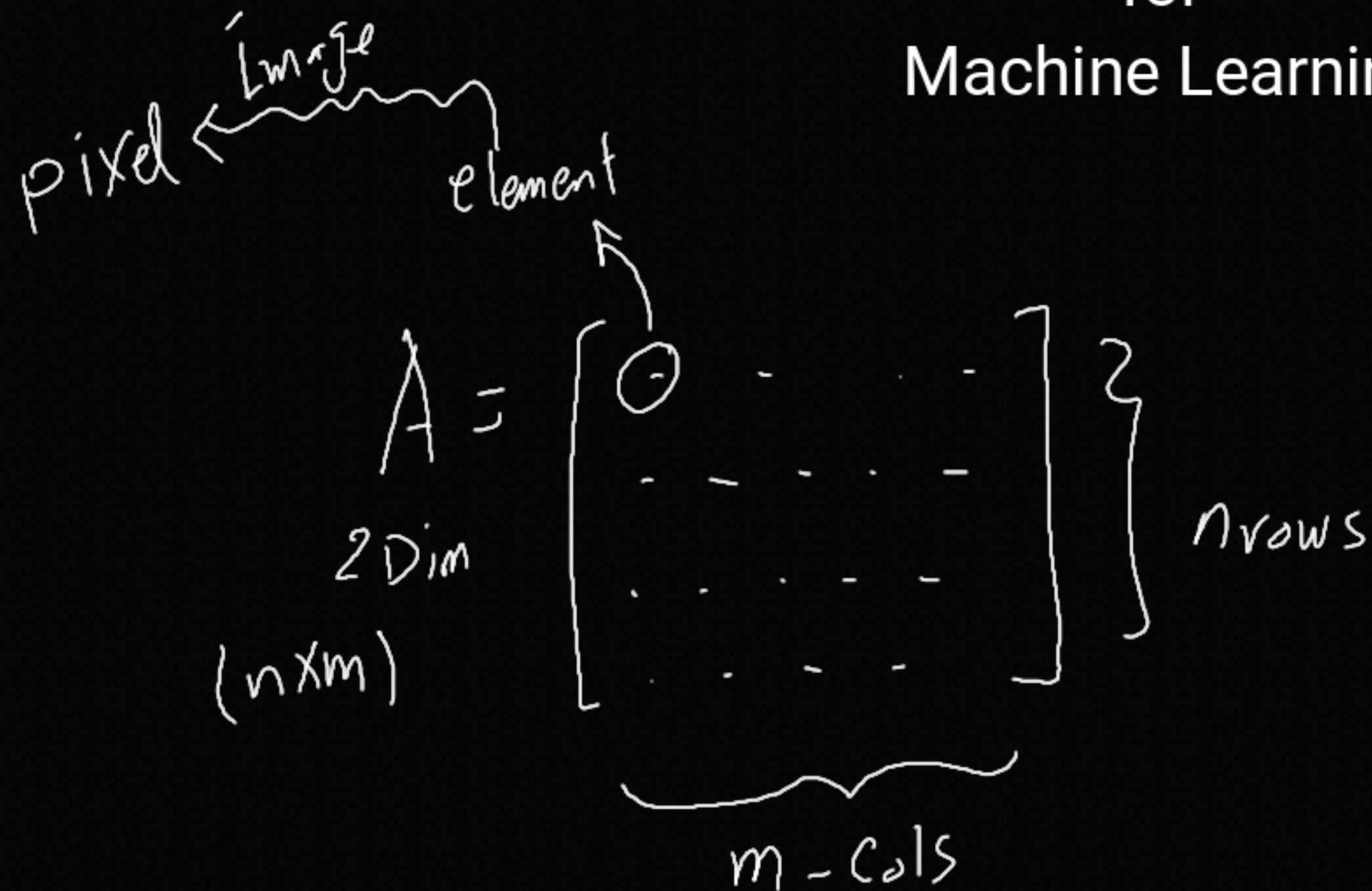
$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1m} \\ a_{21} & \dots & \dots & \dots & a_{2m} \\ \vdots & & & & \\ a_{n1} & a_{n2} & \dots & \dots & a_{nm} \end{bmatrix} \left. \vphantom{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1m} \\ a_{21} & \dots & \dots & \dots & a_{2m} \\ \vdots & & & & \\ a_{n1} & a_{n2} & \dots & \dots & a_{nm} \end{bmatrix}} \right\} \begin{matrix} n \times m \\ n\text{-rows} \end{matrix}$$

~~Programming~~

Index start at $[0]$

$m\text{-cols}$

Intro to Linear Algebra for Machine Learning



	Age	# Experience	Salary
0	20	1	1000
1	30	2	2000
2	25	3	i
⋮	⋮	⋮	⋮
1000	⋮	⋮	⋮

data(1000, 3)

1000 rows
3 cols

(2D)

$$A \begin{bmatrix} a_{00} & \dots & a_{0m} \\ \vdots & a_{ij} & \vdots \\ a_{n0} & \dots & a_{nm} \end{bmatrix}$$

$i = \text{row}$
 $j = \text{col}$

← one_row

one element

$[5]$

(1×1)

↓

Scalar

0D

↑

one_col

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} (4 \times 1)$$

$$[1 \ 2 \ 3 \ 4]$$

(1×4)

Vector

(1D)

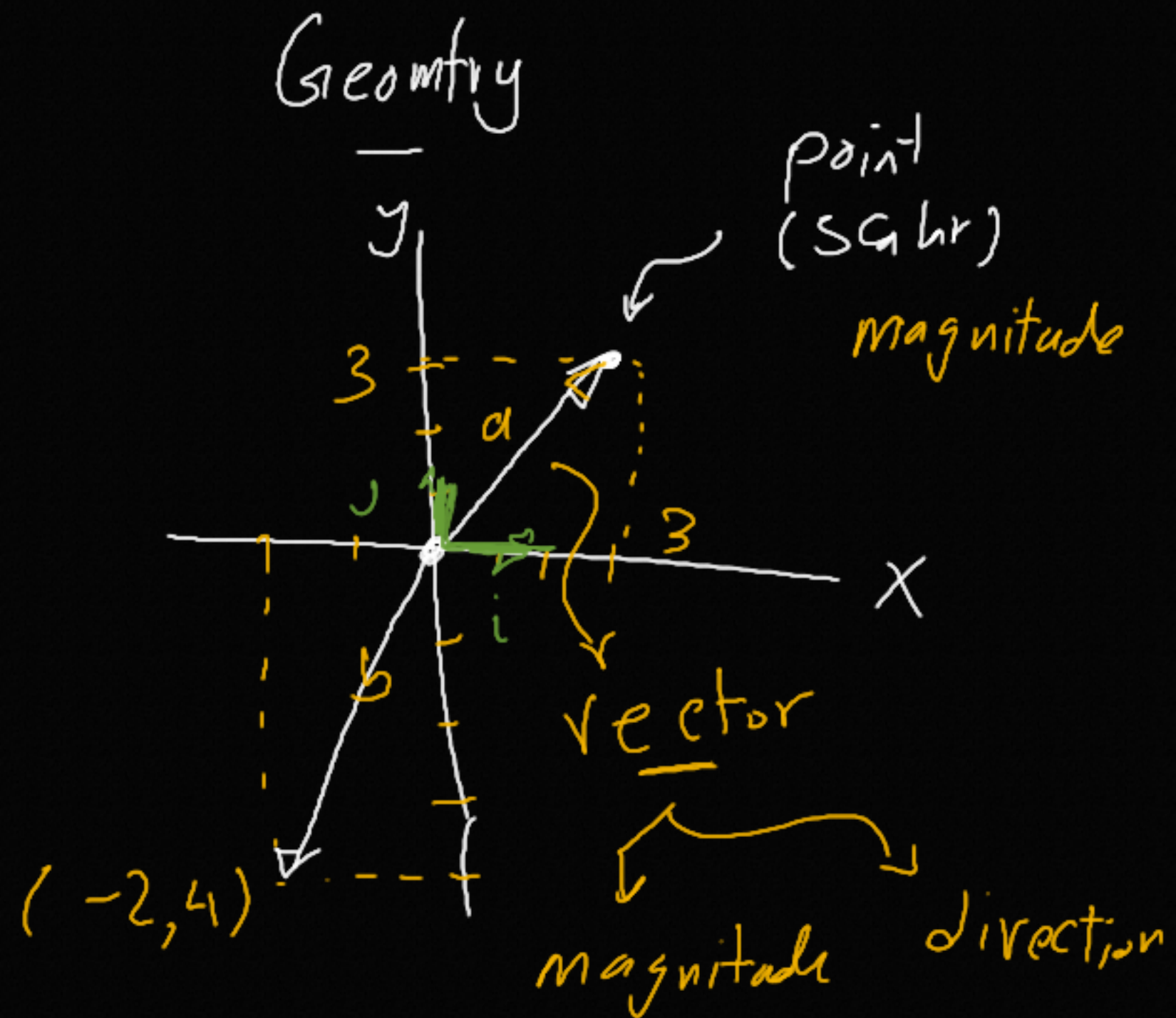
row vector $\rightarrow$ $\begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$ matrix
 column vector $\downarrow$

$$i = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$j = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$a = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$b = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$



Intro to Linear Algebra for Machine Learning

Element by Element operations $\Rightarrow$ Same Dimension

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

2×3

$$B = \begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \end{bmatrix}$$

2×3

Add

$$C = A + B$$

2×3 2×3 2×3

$$= \begin{bmatrix} 11 & 22 & 33 \\ 44 & 55 & 66 \end{bmatrix}$$

Suh

$$D = A - B$$

$2 \times 3 \quad 2 \times 3 \quad 2 \times 3$

$$= \begin{bmatrix} -9 & -18 & \dots \\ \dots & \dots & \dots \end{bmatrix}$$

$\vec{v}_1 =$

3×1

$=$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\vec{v}_2 =$

4×1

$=$

$$\begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}$$

$\vec{v}_3 =$

3×1

$$\begin{bmatrix} -1 \\ -3 \\ -5 \end{bmatrix}$$

Multiplication / Division (Element by element)

$$A / B = \checkmark$$

$\Downarrow$
Same dimension

$$A * B = \checkmark$$

with scalars

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad \begin{matrix} \textcircled{\checkmark} \\ + \\ - \\ * \\ / \end{matrix} \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

$$A + (10)$$

$2 \times 3 \quad 1 \times 1$